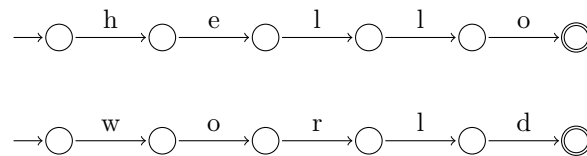


Before starting, make sure you **read all the questions** and understand what is asked of you. Write your answers to the questions **in the exam sheet** and make sure you **justify all of your answers in a separate sheet of paper**. Multiple-choice questions (where options are lettered) have a **unique correct answer**; questions with checkboxes **may have multiple boxes which are true** and you must be exhaustive in marking all the ones that are true.

For the first two questions below, we will focus on the languages accepted by the following finite-state automata. We refer to the one above as $\mathcal{A}_H$ and the one below as $\mathcal{A}_W$; their corresponding languages are $\mathcal{L}_H$ and $\mathcal{L}_W$.



1. (4 points) Which of the following statements hold true?

- ☒ $\mathcal{A}_H$ is deterministic.
- ☒ $\mathcal{L}_H \cup \mathcal{L}_W$ is a regular language.
- ☒ $\mathcal{L}_H \cup \mathcal{L}_W$ is a context-free language.
- ☒ $\mathcal{L}_H \cup \mathcal{L}_W$ is a context-sensitive language.

2. (8 points) Consider the language $\mathcal{L}$ of sequences of words from $\mathcal{L}_H$ followed by even more words from $\mathcal{L}_W$. Formally, $\mathcal{L}$ is defined as $\{w_1 \dots w_m z_1 \dots z_n \mid w_i \in \mathcal{L}_H, z_j \in \mathcal{L}_W, m \leq n\}$. Now, which of the following statements about the language $\mathcal{L}$ hold true?

- ☐ The language is finite.
- ☐ It is a regular language.
- ☒ It is a context-free language.
- ☒ It is a context-sensitive language.

3. (12 points) According to the “longest match” principle (a.k.a. maximal munch), how many tokens does a scanner simulating the automaton below produce when reading the following?

- (a) *abbac*

Solution: 2, namely *abb|ac*

- (b) *bbcbcaab*

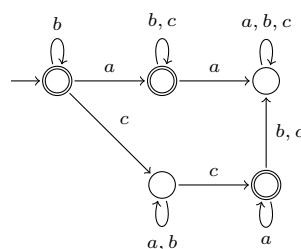
Solution: 2, that is *bbcbcaa|b*

- (c) *ccaaabaa*

Solution: 3, namely *ccaaa|ba|a*

- (d) *becabac*

Solution: 2 again, because *bcba|bac*



4. (6 points) Suppose you are tasked with making a compiler for a new programming language with scoping. Which ones from the following list are valid ways of arranging your information about declared variables?

- A single symbol table with a column for the scope context.
- Multiple symbol tables.
- A tree of symbol tables.

5. (4 points) The control-flow graph is a directed graph with information about the order in which instructions from a program can be executed. Based on how branching is abstracted, the CFG allows us to: A. find instructions that can be reached, B. find instructions that cannot be reached, or C. both.

6. (10 points) Does the following listing contain valid LLVM IR or does it have a bug? Circle the line number of any line containing bugs.

```

1      ; set %tmp iff %a > %b
2      %tmp = icmp sgt i32 %a, %b
3      ; goto iftru if $tmp is set
4      ; goto iffls otherwise
5      br i1 %tmp, label %iftru, label %iffls
6 iftru: %c = 1
7      ; goto end
8      br label %end
9 iffls: %c = 2
10     br label %end
11 end:

```

Solution: I am expecting line 9 will be circled (or both 6 and 9). This is because the same register cannot be assigned twice. Indeed, LLVM IR is a static single assignment intermediate representation.

7. (16 points) Compute the sizes of $\text{First}^1(N)$ and $\text{Follow}^1(N)$, for each nonterminal N , based on the following grammar with terminal set $T = \{a, b, c, d, \$\}$.

(1)	S	$\rightarrow$	$A\$$
(2)	A	$\rightarrow$	$aCbD$
(3)		$\rightarrow$	DAb
(4)		$\rightarrow$	ab
(5)	C	$\rightarrow$	c
(6)	D	$\rightarrow$	d

Nonterminal	Size of $\text{First}^1(\cdot)$	Size of $\text{Follow}^2(\cdot)$
S		
A		
C		
D		

Solution:

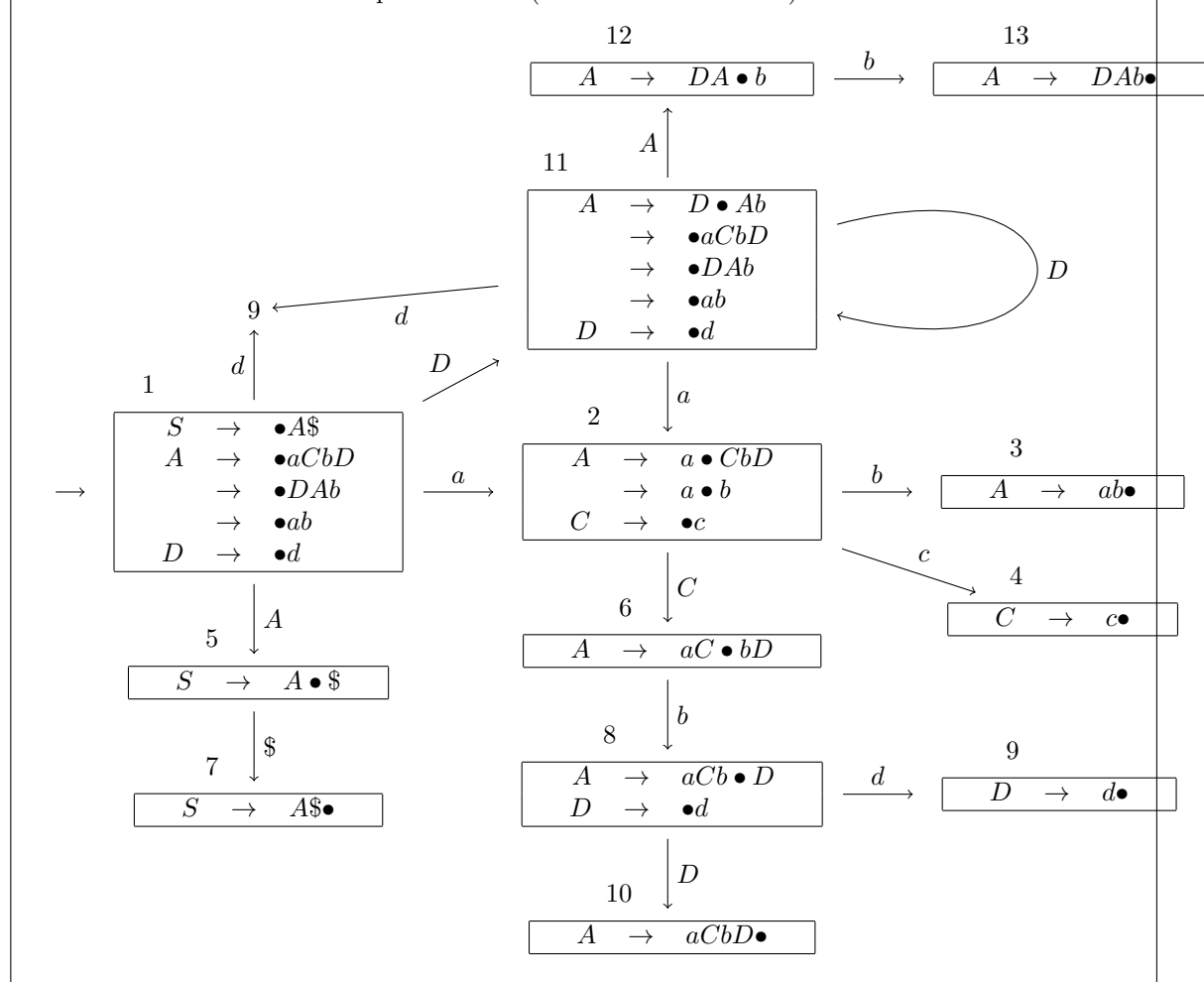
Nonterminal	First	Follow
S	a, d	$\emptyset$
A	a, d	$\$, b$
C	c	b
D	d	$a, d, \$, b$

8. (10 points) Concerning the grammar from the previous question, which of the following statements (if any) holds true?

- ☐ The grammar is LL(1).
- The language is LL(1).

9. (10 points) How many states does the canonical finite-state machine for the grammar from the previous (two) questions have?

Solution: The machine is depicted below (with states numbered). It has 13 states.



10. (10 points) Concerning the grammar from the previous question, which of the following statements (if any) holds true?

- The grammar is SLR(1).
- The language is SLR(1).

Solution: From the canonical finite-state machine it is clear that the language is LR(0). Hence, it is SLR(1) too.

11. (10 points) Give an example of a context-free language that cannot be recognized by a deterministic push-down automaton.

Solution: There are multiple examples, but a common one is the language of palindromes since one has to guess when to start popping instead of pushing into the stack. (Note that palindromes split by a special character is in fact a deterministic PDA language, so it has to be the one without delimiter.)